F9

Predicting Falcon 9 Booster Landings

A data science case study using public SpaceX launch data

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github.com/Sumant-crty/WinningSpaceRaceWithDataScience

Reusing the booster is what makes Falcon 9 cheap

$62M

Falcon 9 launch price

$165M+

typical price from other providers

Most of the price gap comes from recovering and reusing the first stage. Not every booster is recovered: some crash, and some are deliberately expended because of payload or orbit.

The question

Will the first stage land?

If we can predict landing success, a competitor can estimate SpaceX's cost for a launch and bid against it.

A five-step pipeline from raw API data to a model

1

Collect

SpaceX REST API and Wikipedia scraping

requests, BeautifulSoup

2

Clean

Falcon 9 only, imputed payload, binary landing label

pandas

3

Explore

SQL queries and charts by site, orbit, payload and year

SQLite, Seaborn

4

Visualize

Launch-site maps and an interactive dashboard

Folium, Plotly Dash

5

Model

Four classifiers tuned with 10-fold cross-validation

scikit-learn

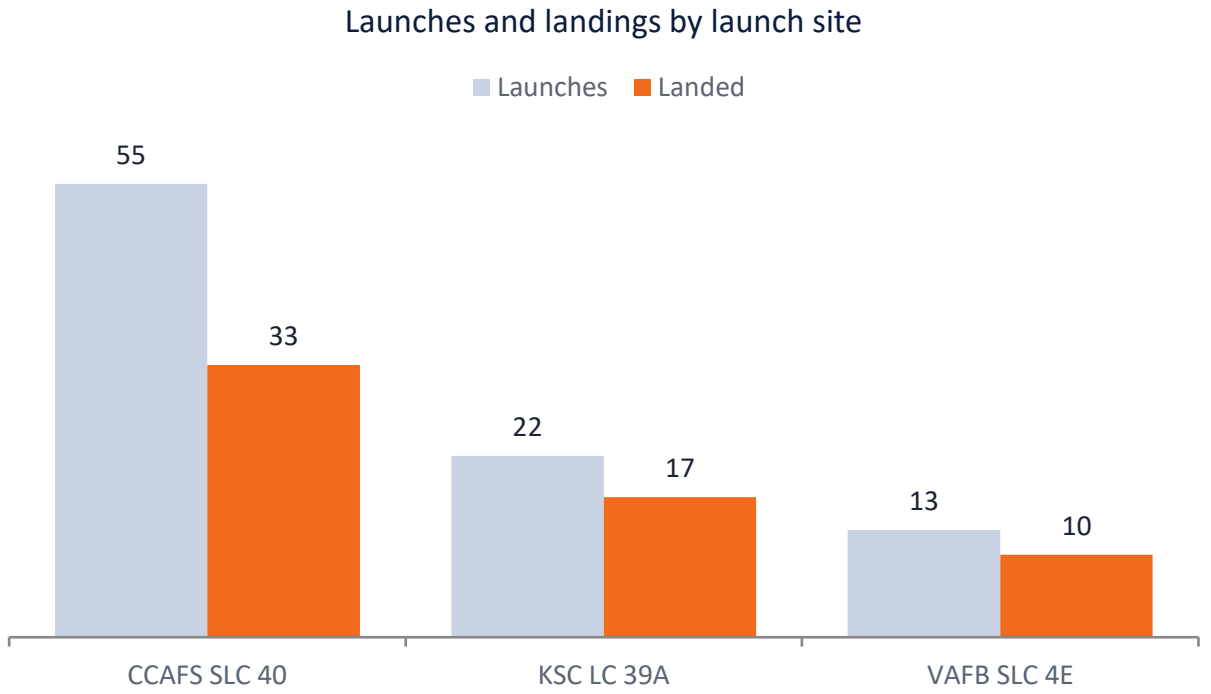
90 launches, and two out of three boosters landed

90

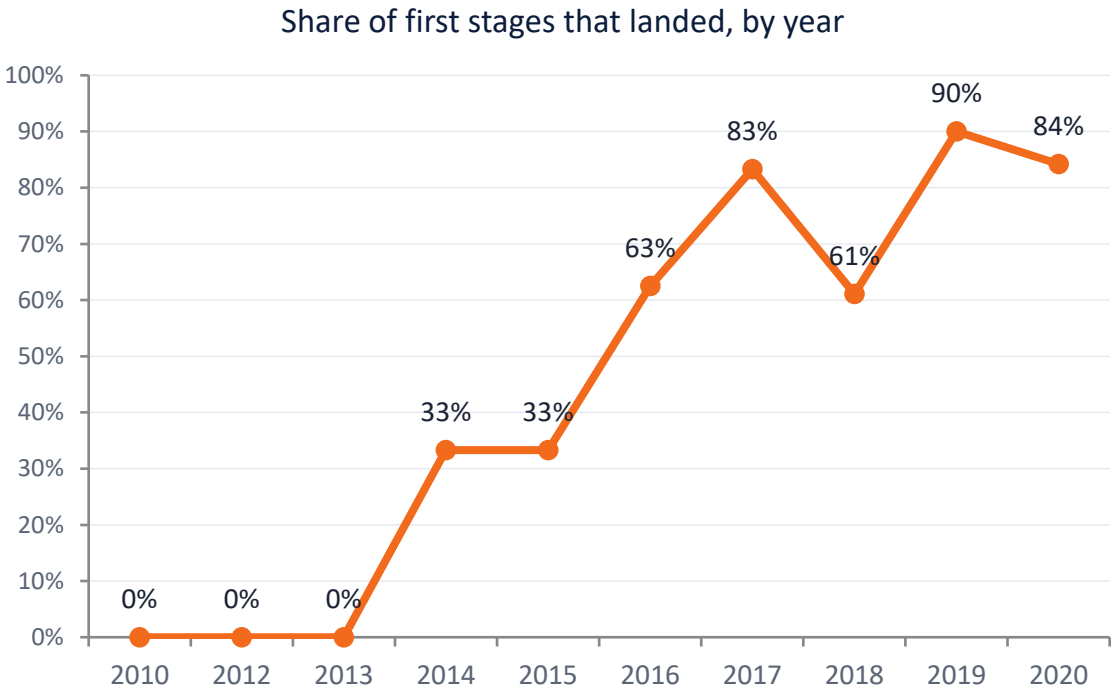
Falcon 9 launches (2010–20)

66.7%

first stages landed



Experience is the strongest landing signal



36%

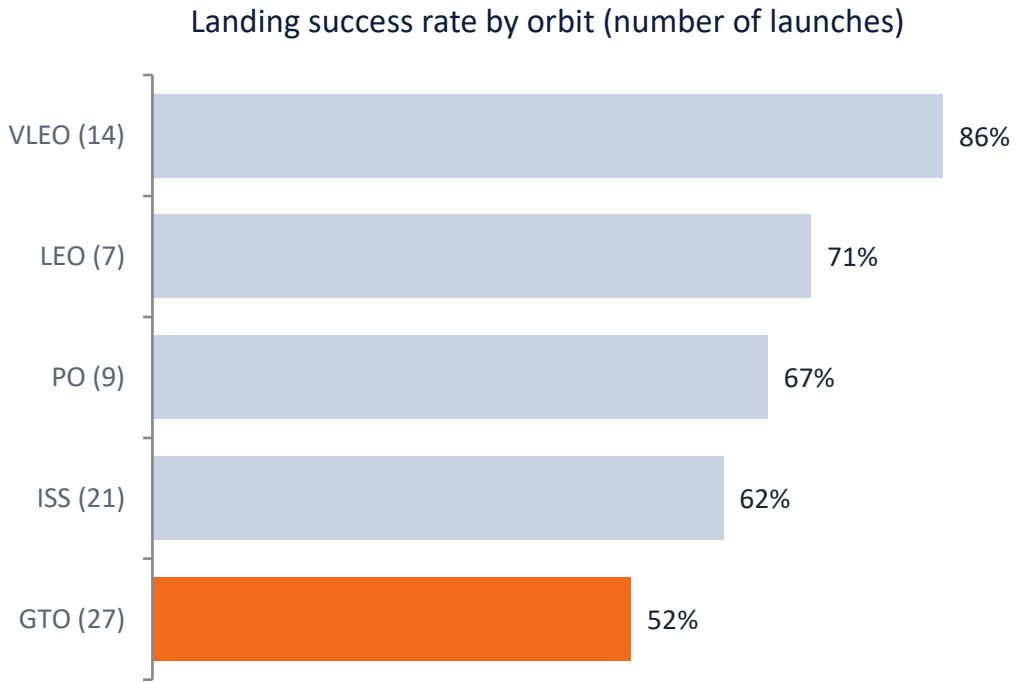
landed in the first 25 flights

78.5%

landed in the 65 flights after that

2018 dipped to 61% before reaching 90% in 2019.

Orbit matters: GTO missions land least often

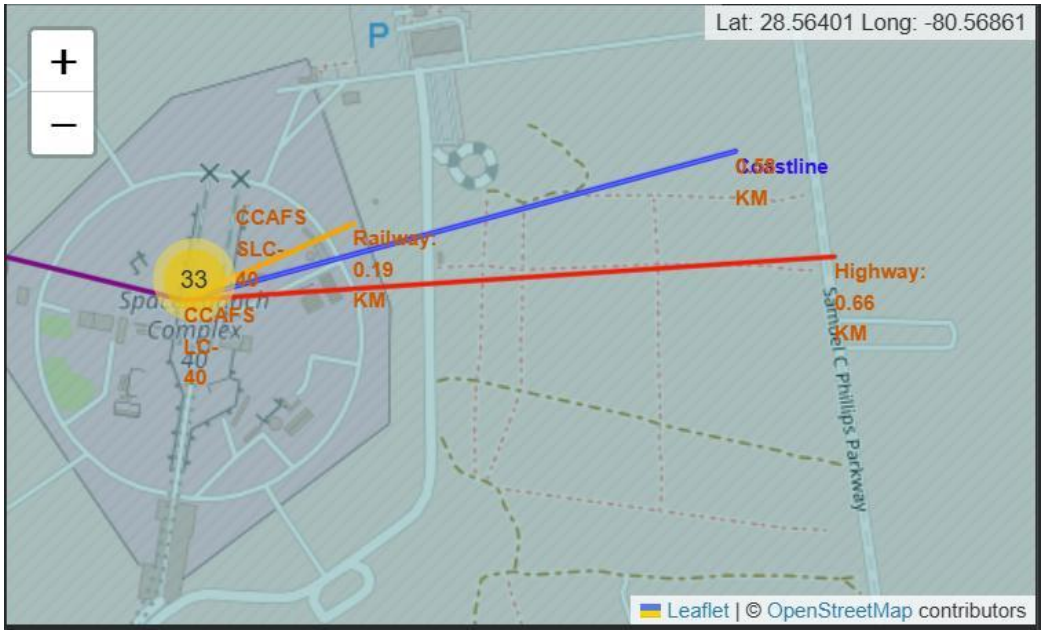


GTO is the most common orbit and the hardest to land from: heavy payloads to a high-energy orbit leave little fuel for the booster's return.

For Polar, LEO and ISS missions, heavy payloads still land successfully most of the time.

Orbits with fewer than 5 launches (ES-L1, GEO, HEO, MEO, SO) are left out: their 0% or 100% rates rest on 1–3 launches.

Launch sites hug the coast and transport links



0.19 km

to a railway

0.58 km

to the coastline

0.66 km

to a highway

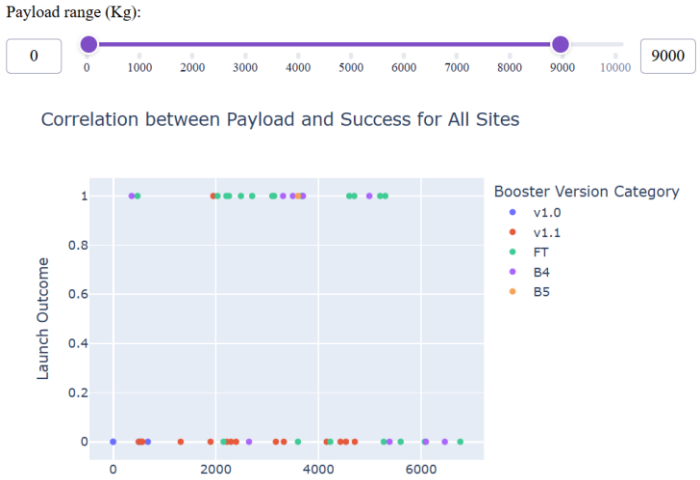
23 km

to the nearest city (Titusville)

Distances from CCAFS LC-40, measured on the Folium map

A dashboard to explore sites and payloads

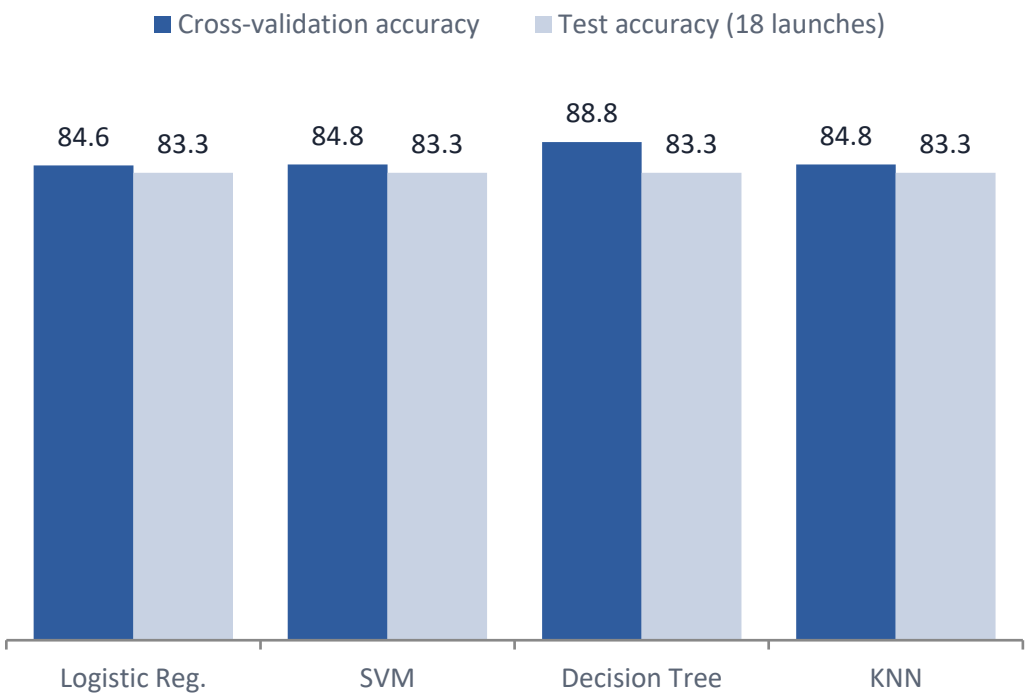




KSC LC-39A has the highest success rate (76.9%) and the largest share of successful launches (41.7%).

Payload slider and site filter show how landing outcome varies with payload mass and booster version (Plotly Dash).

All four models reach 83% test accuracy



Confusion matrix (all four models)

12 landed, predicted landed	**0** landed, predicted failure
3 failed, predicted landed	**3** failed, predicted failure

Every landing is caught; the only errors are 3 false positives.

What these results can and cannot tell us

Limitations

- Only 90 launches; the test set has 18, so one prediction moves accuracy by 5.6 points
- All models tie on the test set, so the ranking rests on cross-validation alone
- Data ends in 2020; landing reliability has changed since

Next steps

- Repeated, stratified cross-validation to compare models fairly
- Add recent launches from the SpaceX API
- Engineer features such as booster reuse count, block version and landing type

Key takeaways

1. **Experience** Landing success rose from 36% in the first 25 flights to 78.5% afterwards

2. **Orbit** GTO missions land least often (52%); VLEO and LEO most often

3. **Model** 83% test accuracy with four classifiers; more data is needed to choose between them

Code, notebooks and dashboard: github.com/Sumant-crty/WinningSpaceRaceWithDataScience